

Research Data Request — Anonymized Poker Hand Histories

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Dissertation topic: Adaptive strategy learning in multi-agent imperfect-information environments — AI agents that infer opponent behavior in real time and exploit it *safely*. The PhD is individual and independent of the company; this request is for anonymized research data only.

The request — 30-second version

- **On the order of 30–100 million anonymized cash-game hands** — the upper end is a comfortable ceiling, not a requirement (volume analysis in Details).
- **Player diversity beats volume:** the ideal shape is roughly **2,000–5,000 distinct players with 20–30k hands each** — the research studies how fast an agent can profile a new opponent, which needs many distinct players more than deep individual histories.
- **One consistent format:** single table size (6-max preferred), cash game, narrow stakes band.
- **Stable pseudonymous player IDs** — the same ID across all of a player’s hands. No real identities, names, or account data needed, ever.
- **Per hand:** full action sequence with exact bet sizes, positions, starting stacks, board cards, timestamp.
- **High-value bonus if logged:** hole cards for *all* hands the platform knows them for (including mucked at showdown and folded hands, if server-side logs exist). This removes the largest statistical bias in hand-history research — see Details.

Commitments up front: I will sign any NDA the company requires. The data stays on a single encrypted machine, is never shared with third parties, and is deleted on request. Only aggregate statistical results would ever appear in publications — and only after company review and approval. The company can be acknowledged or remain entirely unnamed, at its choice.

What the research does with it

Three parts of the dissertation are currently limited to synthetic data; real hands upgrade each one:

1. **Human behavioral prior.** Train a policy network by imitation learning on real play, then use it as the *anchor* for KL-regularized safe exploitation — the paradigm behind Meta’s human-level Diplomacy agent (Science, 2022), not yet applied to poker in the literature.
2. **Opponent modeling from few observations.** Algorithms that infer a player’s type and tendencies within tens of hands — validated against a real player population instead of scripted bots, which is the difference between a toy result and a credible one.
3. **Evaluation on real behavioral data.** The dissertation’s evaluation-methodology contribution explicitly targets validation on anonymized real-world game logs; this dataset *is* that validation.

Potential value for the company

- The opponent-modeling machinery is the same machinery that detects **anomalous play: bots, collusion, multi-accounting**. Anomaly detection is a stated application track of the dissertation — findings and methods can flow back.
- **Population-level insights** into player pool tendencies as a byproduct of the behavioral analysis.
- Association with **published academic AI research** (with full veto rights over anything that references the company or its data).

Data-handling commitments

Concern	Commitment
Legal	NDA / data-use agreement on the company's terms, signed before transfer
Storage	Single encrypted local machine; no cloud copies, no third-party access
Identity	No re-identification attempts; pseudonymous IDs only
Publication	Aggregate results only; company reviews and approves before any submission
Lifecycle	Data deleted on request or at project end, whichever the company prefers

Details (optional reading)

On the requested volume. The headline figure is deliberately a ceiling; useful research begins far lower. Per player, the common behavioral tendencies (looseness, aggression, positional habits) stabilize statistically within roughly 1,000–2,000 hands, while rarer situational patterns — responses to re-raises, street-by-street betting lines — occur in only a few percent of hands and need on the order of 10,000–20,000 hands per player to estimate reliably. Beyond ~50,000 hands per player, additional depth mainly serves niche studies of how a player's style drifts over time. Across the population, behavioral clustering and rapid-profiling experiments benefit from breadth: about 500 players is the working minimum, 2,000–5,000 is the sweet spot, and past ~10,000 the returns flatten. Combining the two axes, the golden initial dataset is about **2,000–5,000 players with 20,000–30,000 hands each**; volumes above ~100 million hands add storage cost without research gain. If the full ask is impractical, a floor of ~500 players $\times$ 10,000 hands (roughly 5 million total) already supports the core experiments at reduced statistical power. One further economy: a single 6-max hand provides observations for all six seats, so if the players form an overlapping pool, the number of distinct table-hands needed shrinks several-fold.

Research context. The dissertation develops three contributions: (1) a *Behavioral Adaptation Framework* — real-time inference of opponent strategy from observed actions; (2) *Multi-Agent Safe Exploitation* — exploiting weaker opponents while bounding worst-case losses, extending two-player safety theory toward N-player games; (3) an *Evaluation Methodology* for measuring adaptability and robustness. State-of-the-art poker AI (Libratus, Pluribus) plays a fixed equilibrium strategy

identically against everyone; the research gap is principled, safe *adaptation* — and adaptation research is only as good as the behavioral data it is validated on.

Why hole-card completeness matters. Public hand histories reveal hole cards only at showdown (~25–30% of hands), and showdown hands are a biased sample — passive lines reach showdown far more often than aggressive ones. Models trained only on shown-down hands inherit that bias. If the platform’s server-side logs retain dealt hole cards for mucked or folded hands, including them turns a biased 25% sample into a complete unbiased dataset — the single largest quality difference achievable in this kind of research.

Ideal field list per hand. Hand ID; timestamp; table size and stakes; button position; per-seat pseudonymous player ID and starting stack; blinds/antes posted; full ordered action sequence (fold / check / call / bet / raise with exact amounts, per street); board cards per street; hole cards where known; showdown result and pot distribution. Any standard hand-history export format is fine — parsing is my job, not the company’s.

Timeline. No urgency on the company’s side is implied. My infrastructure (parsing, storage, training pipeline) will be ready ahead of time; the data can arrive whenever the legal side is settled.